

WRF Tools case report

Case

Item	Result
Wrfout File	wrfout_d01_2012-05-23_00_00_00.nc
Grid Id	1
Dimensions	{"Time": 25, "south_north": 109, "west_east": 109, "bottom_top": 50, "bottom_top_stag": 51, "soil_layers_stag": 4, "west_east_stag": 110, "south_north_stag": 110}
Selected Point	{"latitude": 54.011009216308594, "longitude": 6.60699462890625, "x": 54, "y": 54}

Quality Control

Item	Result
Errors	0
Warnings	0
Issues	[]

Surface Wind

Item	Result
Mean 10M Wind Speed M S	7.809161186218262
Maximum 10M Wind Speed M S	10.20217227935791
Turbulence Intensity U10 Percent	23.860156110949948

Boundary Layer

Item	Result
Mean Wrf Pblh M	109.13041687011719
Mean Wrf Pblh Excluding Zero M	113.67752838134766
Mean Diagnosed Pblh M	50.75190560121089
Mean Convective Velocity Scale M S	0.0
Stability Class Counts	{"neutral": 1, "stable": 23, "very stable": 1}
Diagnostic Method	First model level at least 0.5 K warmer than the lowest level.
Interpretation	This discrete threshold estimate is constrained by model-level spacing and is not equivalent to WRF PBLH. Negative surface heat flux and predominantly stable classes support a shallow stable boundary layer.

Wind Energy

Item	Result
Rotor Bottom M Agl	40.0
Rotor Top M Agl	200.0
Mean Rotor Equivalent Speed M S	14.315433289574248
Mean Shear Exponent	0.29251918571768715
Mean Veer Degrees	20.262607288502995
Weibull Shape	7.528361997972929
Weibull Scale M S	8.317158239873486
Mean Wind Power Density W M2	319.1913146972656

Spectral And Coherence

Item	Result
U10 Fitted Log Slope	-3.4520581379121436
Integral Time Scale Hours	3.574591193002751
Integral Length Scale M	100492.41168358097
Mean Lateral U10 Coherence	0.951718135044986
Record Length Warning	Only 25 hourly samples: spectra demonstrate the API but cannot resolve an atmospheric inertial subrange.

Thermodynamics

Item	Result
Surface Relative Humidity Percent	88.9725570678711
Maximum Equivalent Potential Temperature K	418.2936096191406
Maximum Brunt Vaisala Frequency S 1	0.06091601774096489
Inversion Level Count First Time	16

Turbulence And Flow

Item	Result
Mean Resolved Tke M2 S2	0.8768419027328491
Mean Shear Production M2 S3	5.8907439779432025e-06
Mean Buoyancy Production M2 S3	1.5078880096552894e-05
Maximum Absolute Vorticity S 1	9.115917782764882e-05
Low Level Jet Detected First Time	True
Jet Core Speed M S	11.770992279052734
Jet Core Height M Agl	260.2864159466699

Precipitation

Item	Result
Domain Mean Total Mm	1.7699477672576904
Domain Maximum Interval Mm	6.484642028808594

Terrain Landuse Resolution

Item	Result
Maximum Terrain Slope Degrees	2.171964594472876
Landuse Category Statistics	{"1": {"count": 1295, "mean": 280.70011040529226, "std": 4.005951712949682, "minimum": 273.103515625, "maximum": 291.3663330078125}, "2": {"count": 93, "mean": 286.2599854828209, "std": 1.434415901748272, "minimum": 2...
Coarsen Regrid Comparison	{"bias": 0.04534820928168068, "mae": 0.2280515728336957, "rmse": 0.32955820089195725, "correlation": 0.9956270063479808}
Temperature Integral Conservation	{"source_integral": 2424524876283904.0, "target_integral": 2424524607848448.0, "relative_error": -1.1071672583183967e-07}
Boundary Difference K	-0.8963462744453636
Vector Rotation Speed Max Error	1.9073486328125e-06

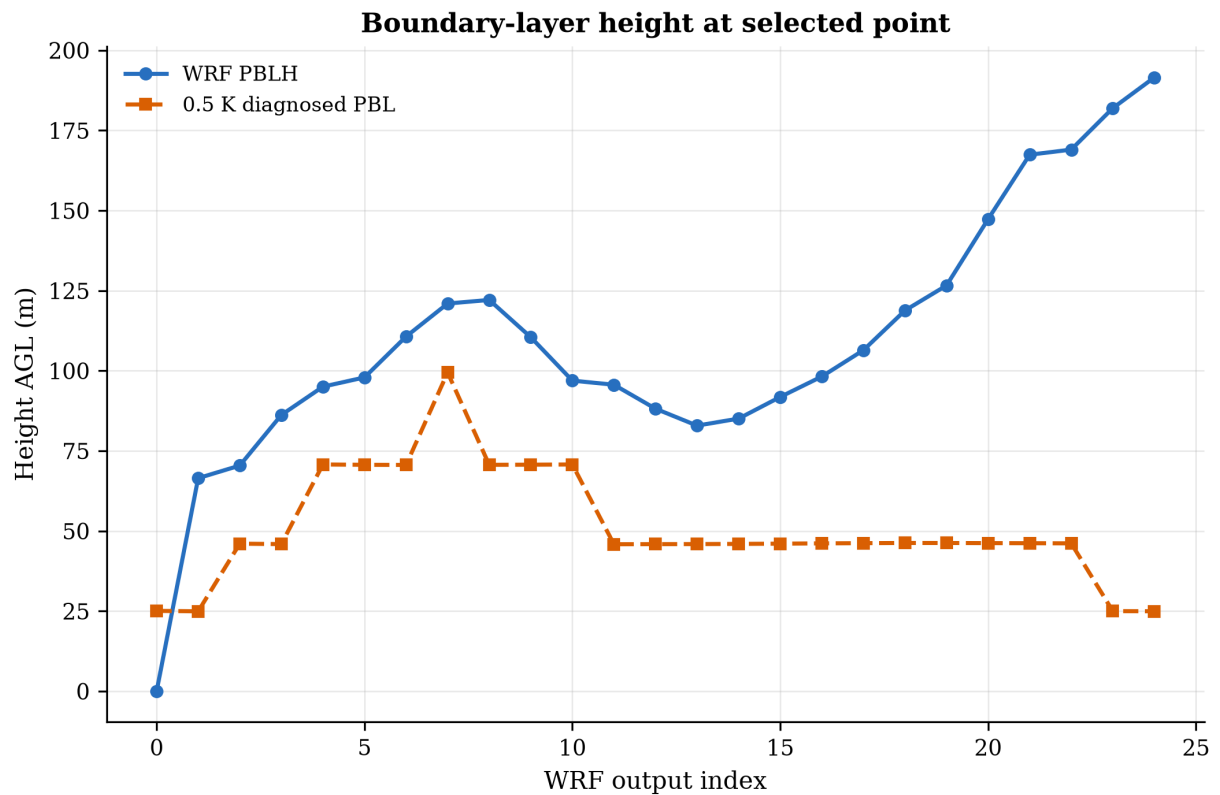
Temporal

Item	Result
Lag1 Persistence	0.9427880774097549
Ramp Threshold M S	0.8283495597844321
Ramp Event Count	1
Minimum Diurnal Mean Speed M S	6.334440231323242
Maximum Diurnal Mean Speed M S	10.20217227935791
Peak Diurnal Hour	22

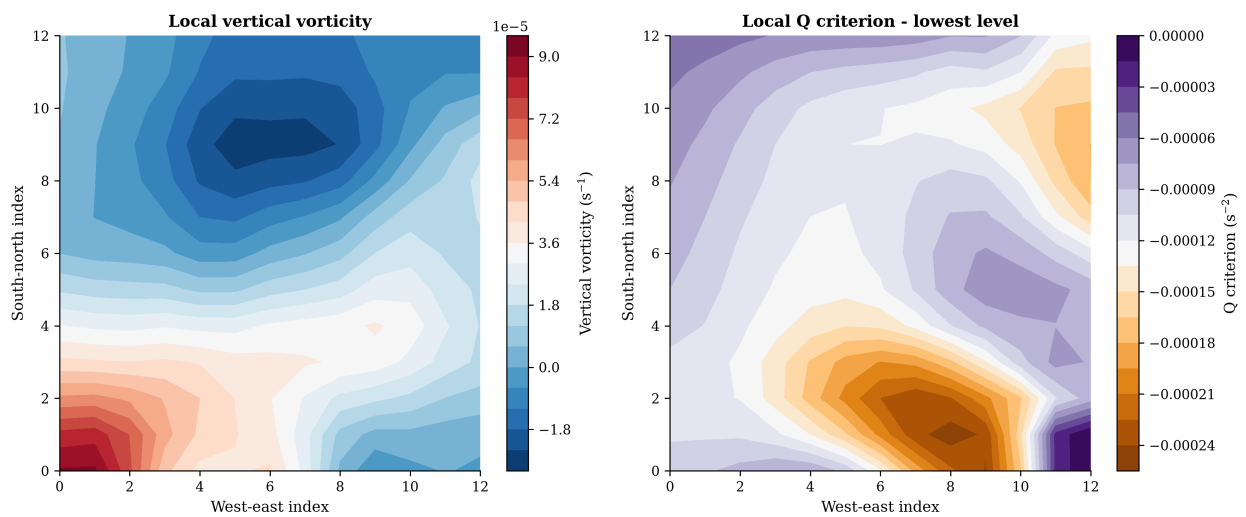
Openfast Conversion

Item	Result
Bts Roundtrip Max Error M S	9.557558688566559e-05
Wnd Roundtrip Max Error M S	0.0015773676025130356

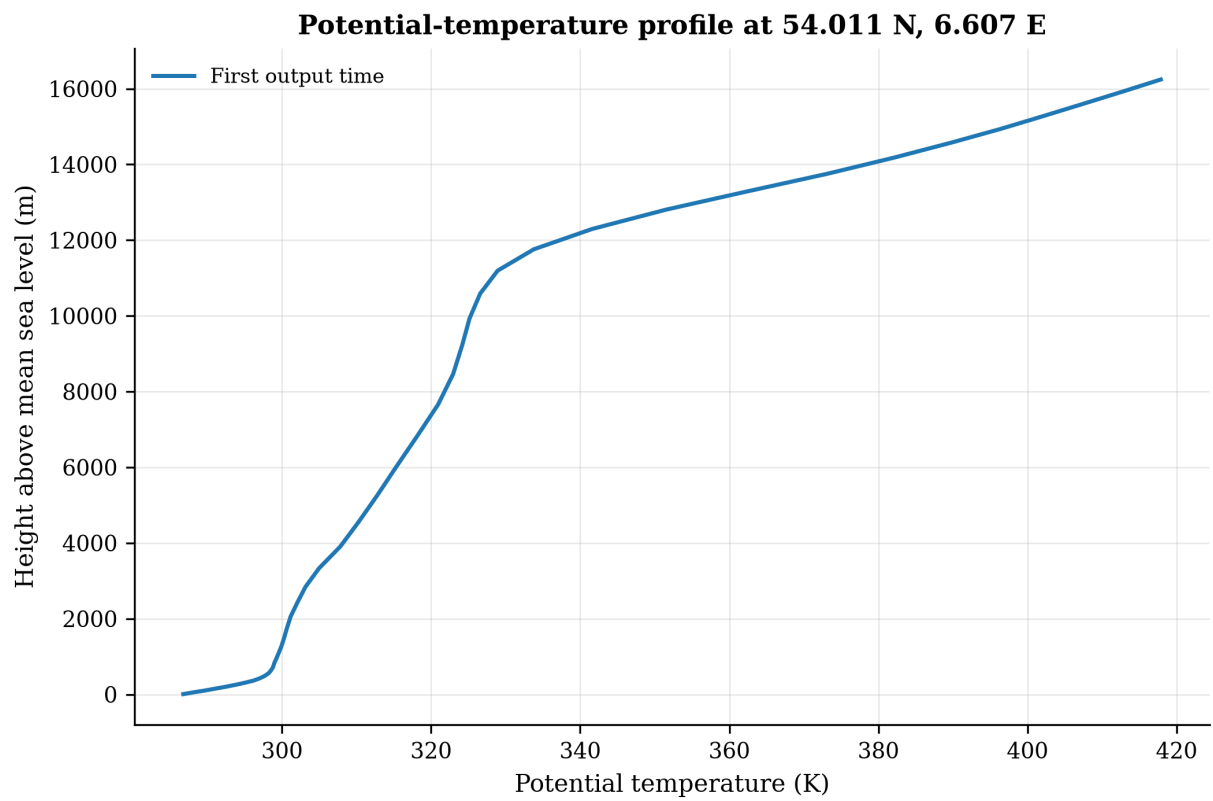
Diagnostic figures



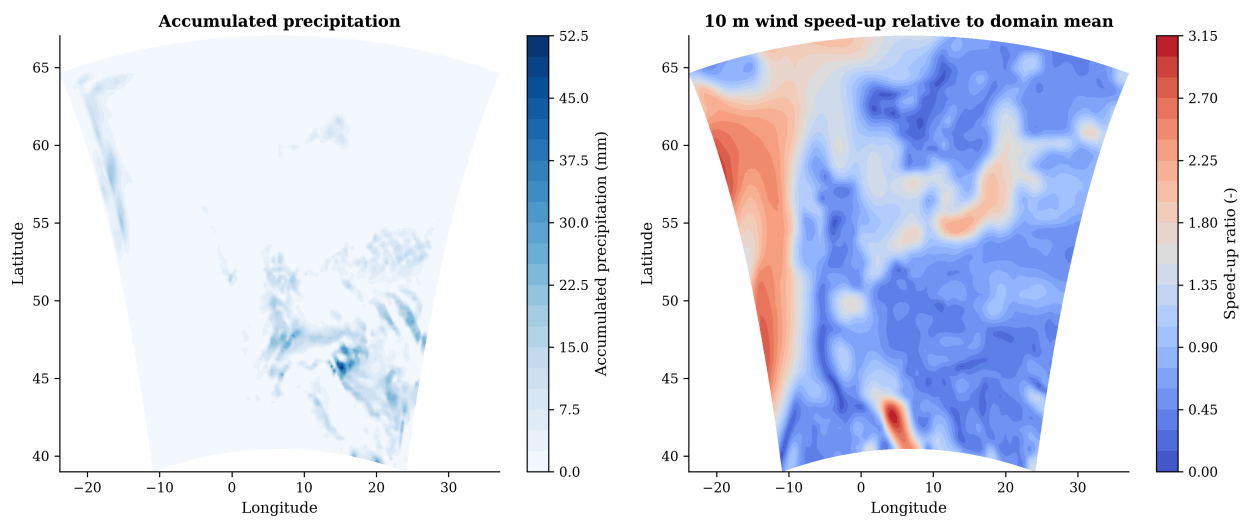
Boundary Layer Height



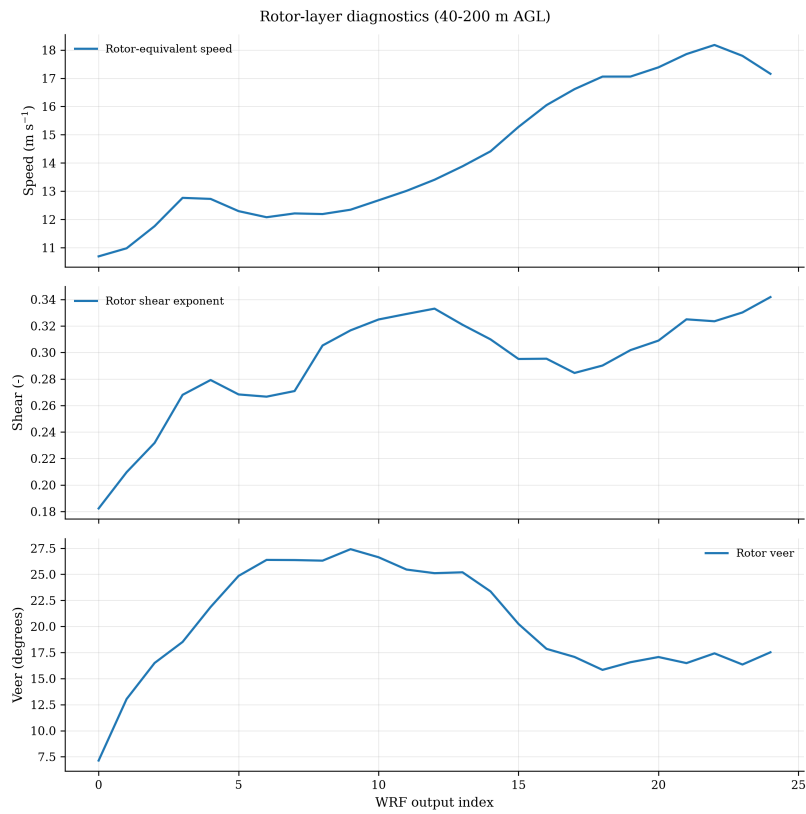
Flow Structure Diagnostics



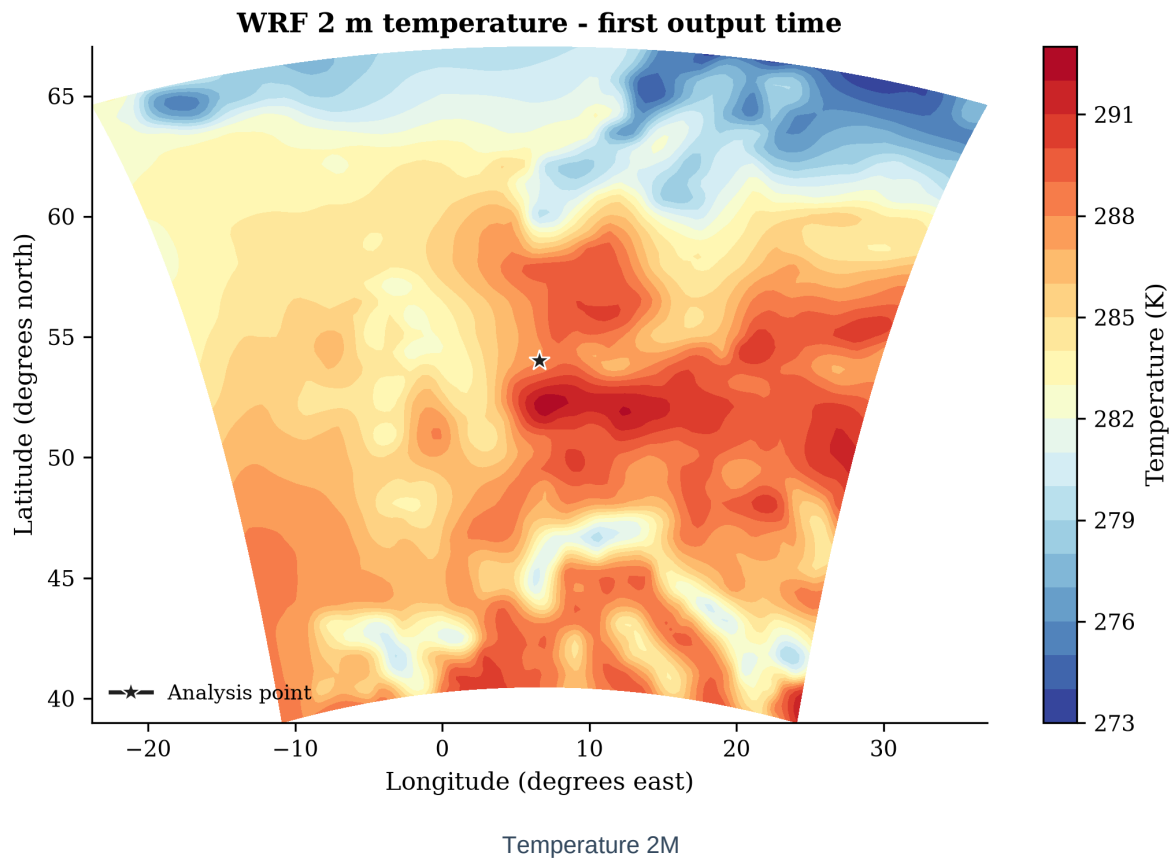
Potential Temperature Profile

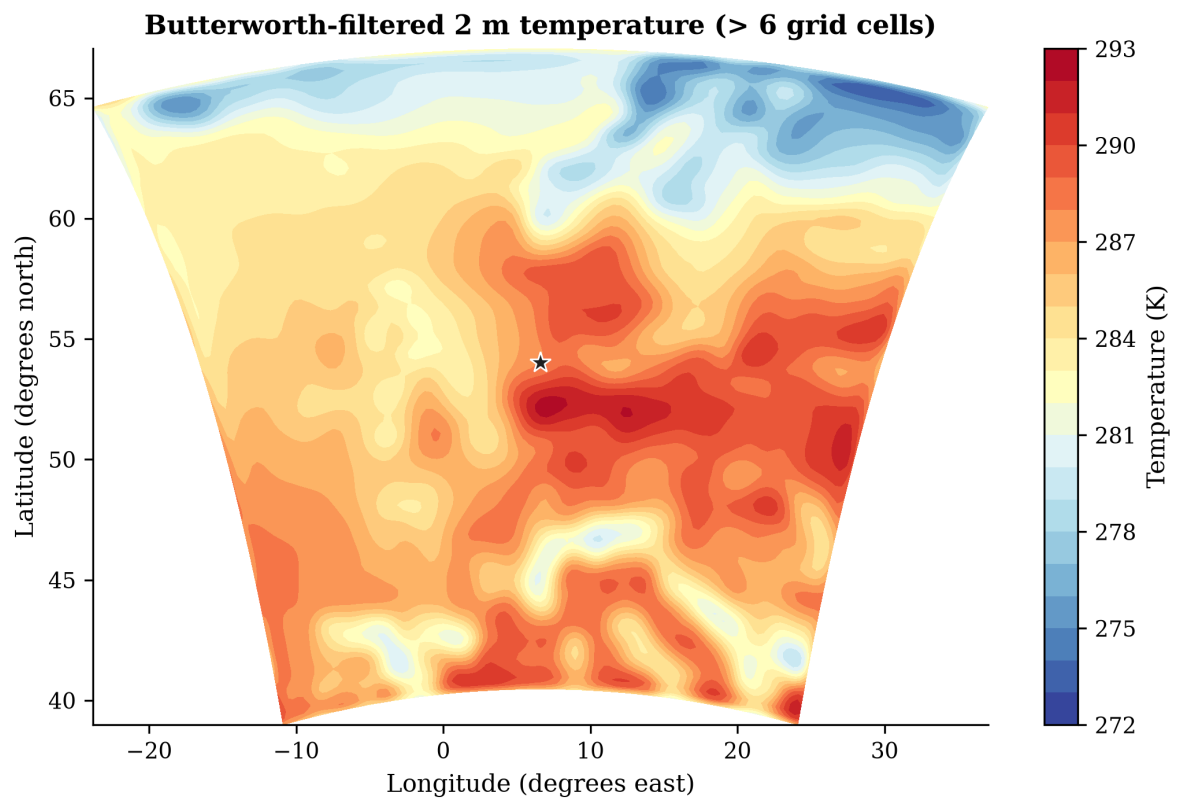


Precipitation And Speedup

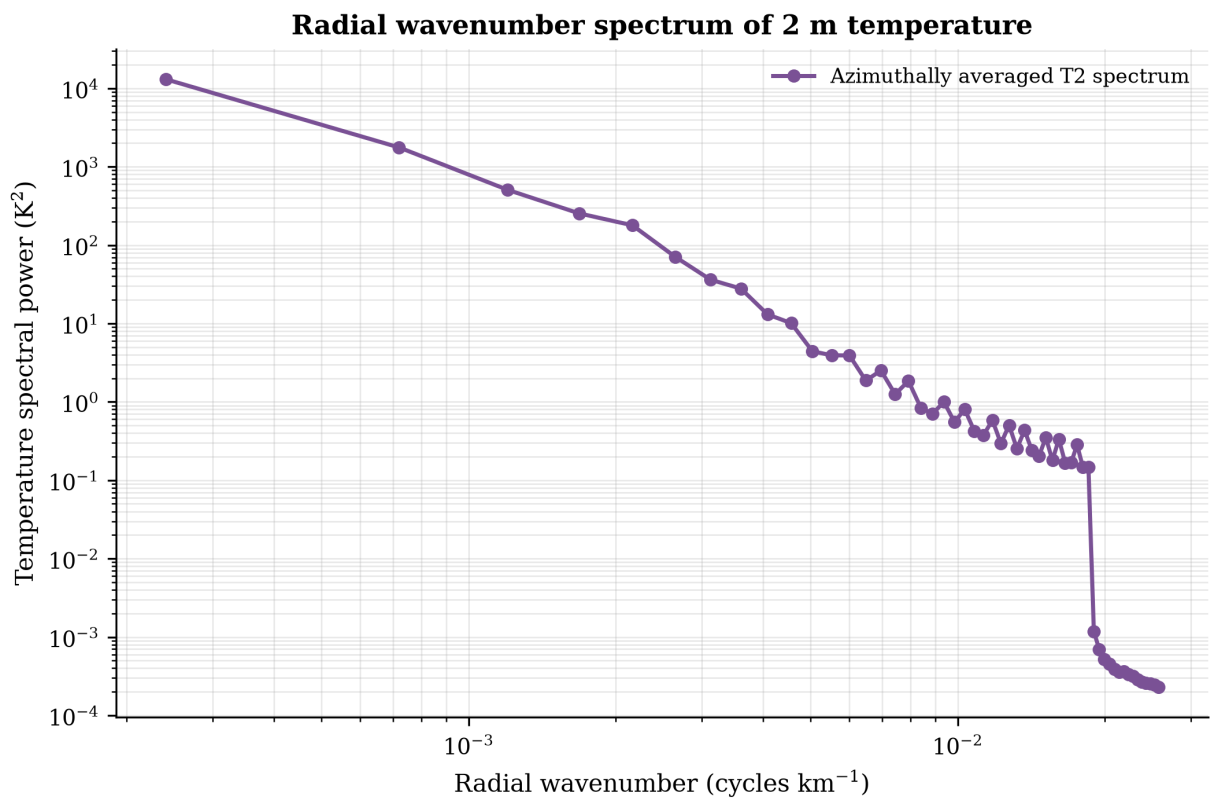


Rotor Diagnostics

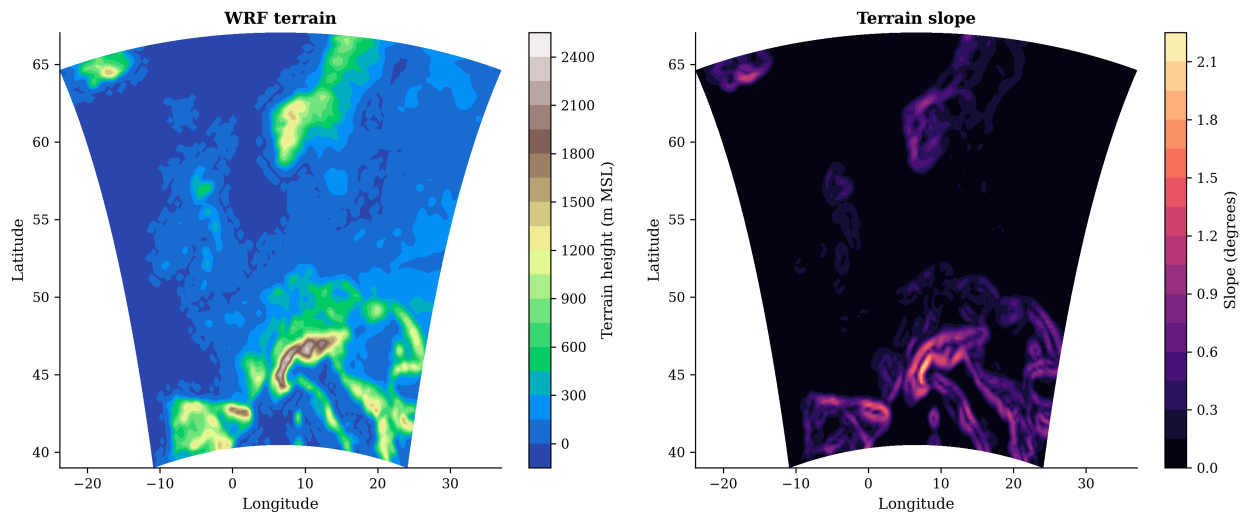




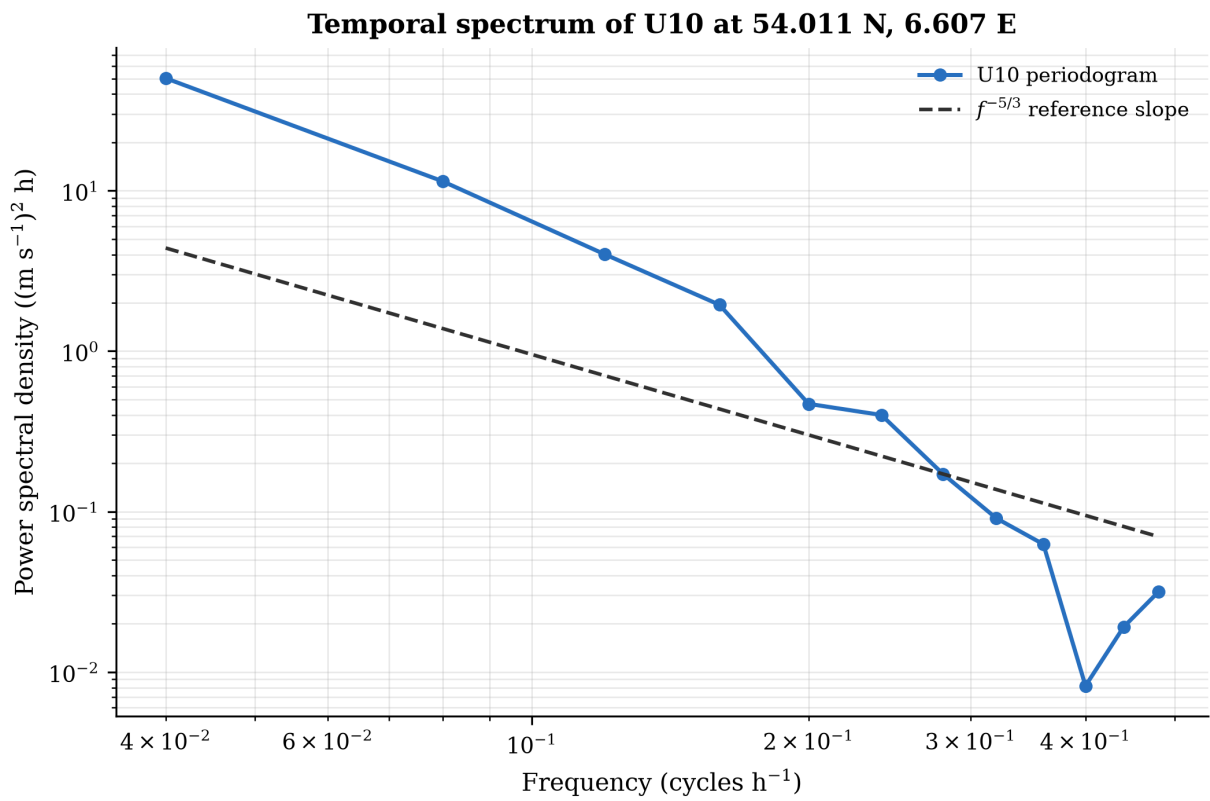
Temperature 2M Filtered



Temperature 2M Wavenumber Spectrum

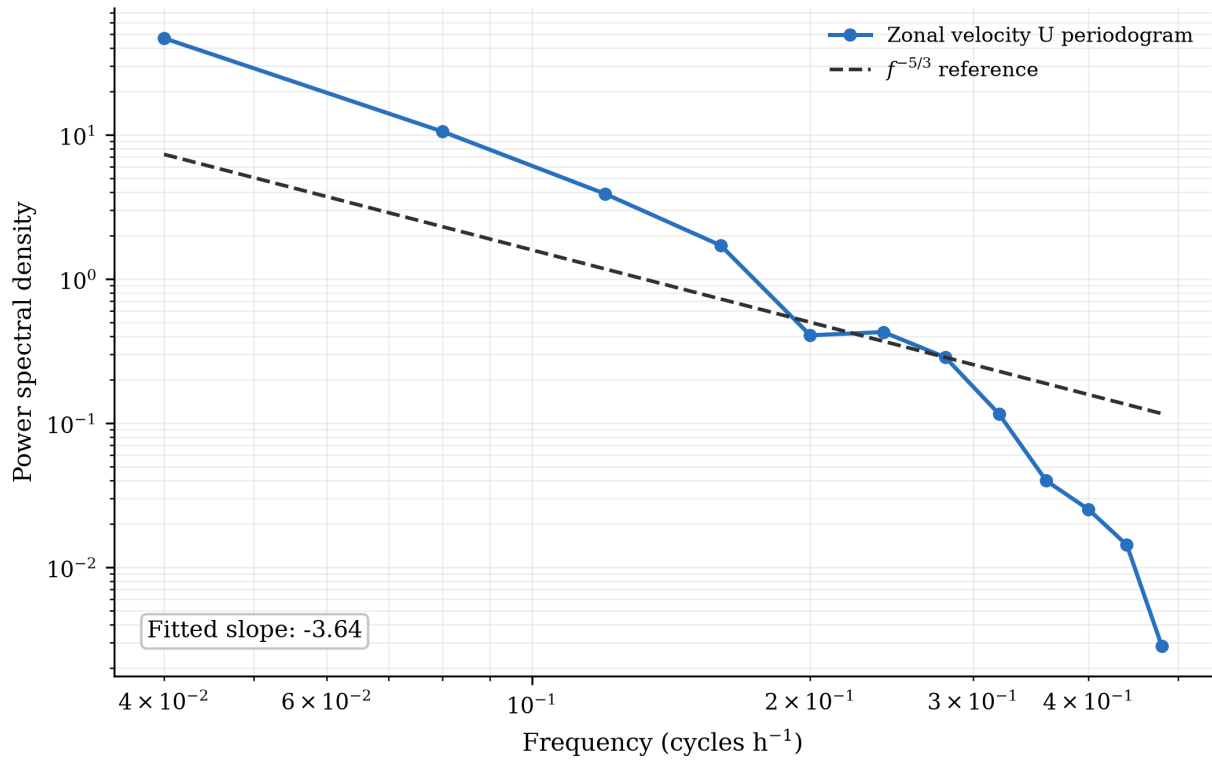


Terrain Diagnostics



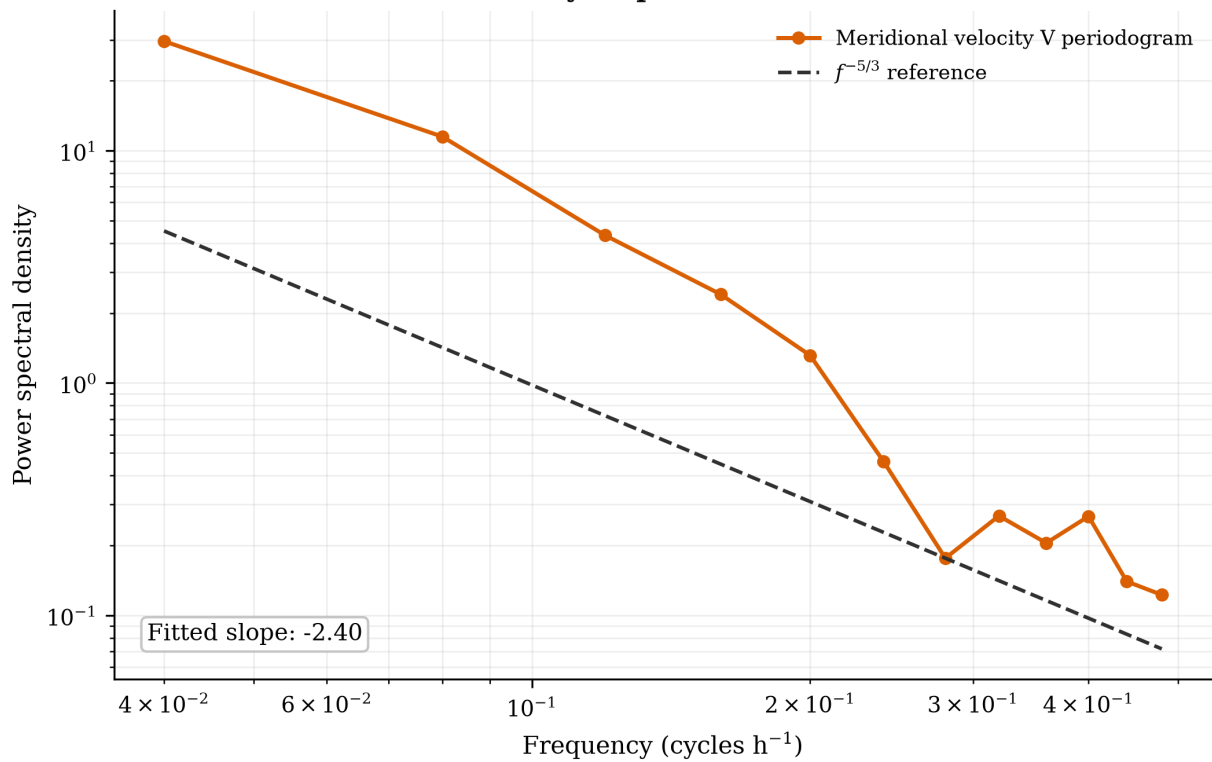
U10 Temporal Spectrum

Zonal velocity U spectrum - first model level

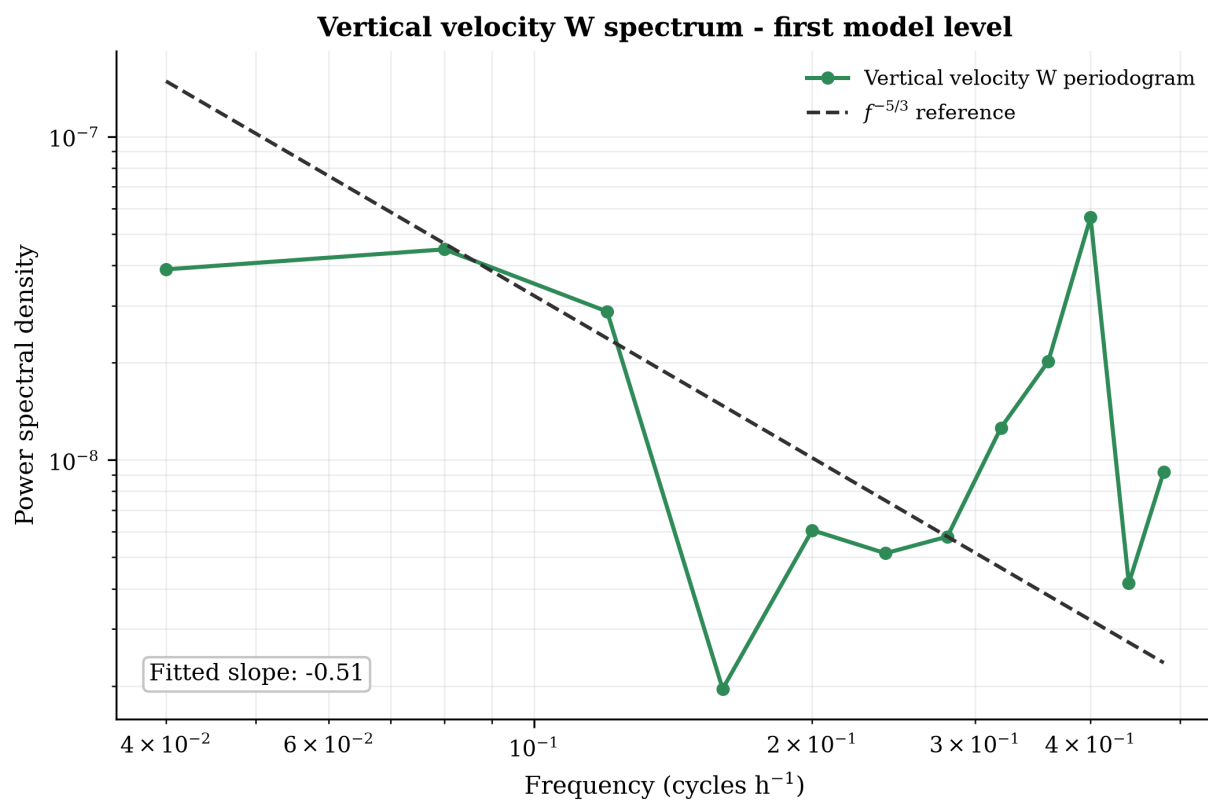


U Wind Spectrum

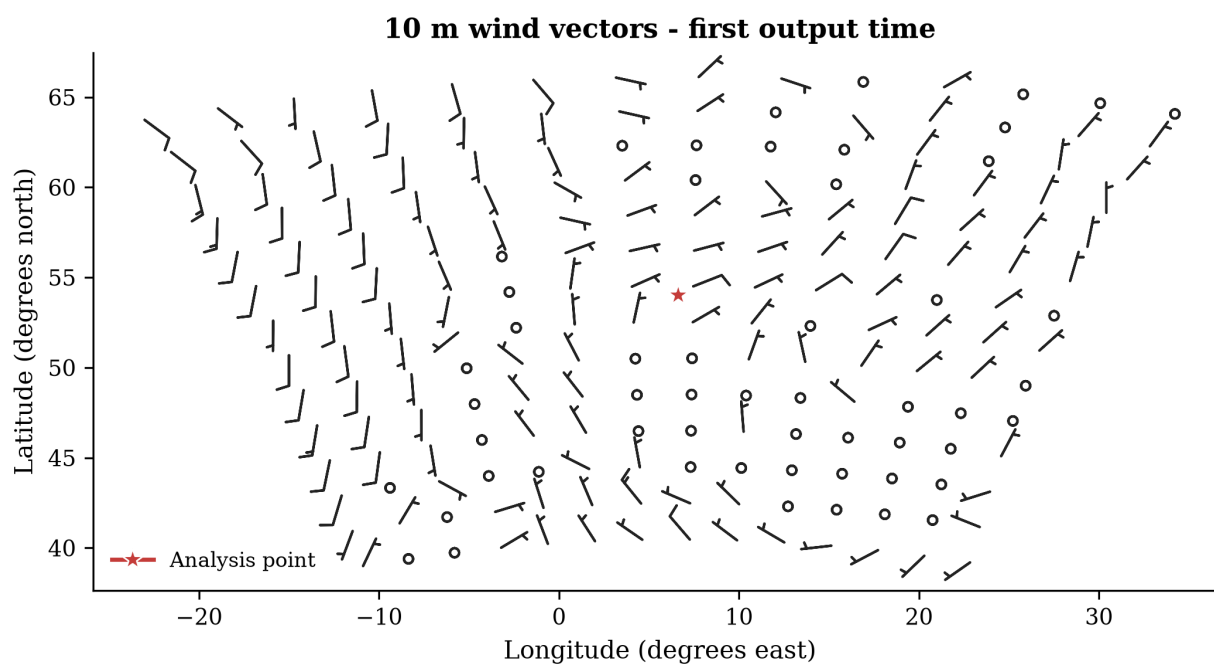
Meridional velocity V spectrum - first model level



V Wind Spectrum



W Wind Spectrum



Wind Barbs 10M

